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Subsea Foundations

Abstract

A subsea foundation comprises an upright pile, a support table on top of the pile, and a capture rod fixed to the pile and extending above a level of the support table. A lower end of the capture rod is fixed to a side wall of the pile, and the capture rod is inclined away from a central longitudinal axis of the pile in an upward direction. A line such as a pipeline or a mooring line is then connected to the subsea foundation by suspending a connecting structure from the line, capturing the connecting structure on the capture rod and pivoting the connecting structure about the capture rod before landing the connecting structure on the support table. The connecting structure can be pivoted about the capture rod around horizontal and/or vertical axes and can be slid down the capture rod before landing on the support table.

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Background/Summary

[0001] This invention relates to subsea foundations for supporting or controlling the position of subsea or floating structures such as pipelines, risers and surface installations used in the offshore energy industry. Embodiments of the invention relate particularly to easing the installation of pipelines or other elongate subsea elements, such as mooring lines, and improving the resistance of subsea foundations to lateral loads.

[0002] In this specification, unless the context requires otherwise, the term ‘foundation’ is not limited just to underlying structures for supporting a downward weight load but also encompasses anchors or other anchorage structures for resisting horizontal or lateral or upward loads, such as tensile loads in mooring lines or in pipelines during pipelay initiation.

[0003] Subsea foundations are often used to stabilise a floating structure. For example, a surface installation such as an FPSO (floating production, storage and offloading vessel) may be moored to the seabed to remain in substantially the same location for many years of oil and gas production. Some taut-leg floating production units are known but they require reinforced anchorages to be installed, which generally requires injection of concrete or grout. More typically, an FPSO is held in place by a mooring system comprising catenary-curved mooring lines arranged in groups to define semi-taut legs. When viewed from above, the mooring lines of each group splay apart to define a spread pattern.

[0004] To resist the predominantly lateral loads applied by catenary moorings, a subsea foundation may comprise a drag anchor that is held in place by its weight and by friction or engagement with the seabed. More commonly, however, anchoring against lateral loads is provided by a pile, this being a vertical element embedded into the seabed.

[0005] Suction piles can withstand upward withdrawal loads better than hammered piles, such as those described in WO 2013/053936, or torpedo piles that are dropped to penetrate the seabed. However, as the lateral resistance of suction piles may not be as strong as that provided by hammered piles or torpedo piles, there is a risk that a suction pile could capsize where lateral loads are high. WO 02/062653 addresses this by connecting a mooring chain to a suction pile beneath the mudline to avoid capsizing the pile under tension in the mooring chain. However, in applications where resistance to upward pulling loads is less significant, suction piles are less favoured than hammered piles or torpedo piles for mooring FPSOs or other surface installations.

[0006] Where a foundation comprises a drag anchor or chain connected to a pile as in WO 02/062653, resistance to lateral movement is strongly directional, being biased to resist in-service tension in a line connected to the foundation. There is much less resistance to lateral movement in opposite or transverse directions, and to upward movement away from the seabed.

[0007] Another use for a subsea foundation is to anchor a pipeline or other elongate element to the seabed. For example, to facilitate pipelay initiation, a subsea foundation may be preinstalled in the seabed before an end of the pipeline is coupled to the foundation. This provides a fixed point for an installation vessel to act against at the start of the pipelaying process. The resulting reaction force keeps the catenary between the vessel and the seabed under tension and avoids longitudinal slippage of the pipeline under that tension.

[0008] A pipeline or other elongate element may also be anchored by a subsea foundation to restrain its movement relative to the seabed in service, for example movement driven by cycles of thermal expansion and contraction. In that case, a pipeline may be anchored not just at an end but

also at an intermediate location along the length of the pipeline.

[0009] A subsea foundation may also be used with a riser that connects a pipeline on the seabed to a surface installation for transporting fluids between them. In that respect, foundations of the invention are primarily concerned with catenary- or wave-configuration risers that are anchored to minimise motion around the touch-down point (TDP) where the riser meets the seabed. For example, a steep-configuration riser is restrained at the TDP by a subsea base structure whereas a pliant-wave riser comprises a subsea anchor that controls motion around the TDP.

[0010] U.S. Pat. No. 7,360,968 takes the approach of combining a structure on a suction pile with a stopper on a pipeline. However, this does not allow for any lateral deviation or even for slight angular deviation from the foreseen path or trajectory of the pipeline.

[0011] In U.S. Pat. No. 7,600,569, a yoke on a pipeline connects to a foundation structure that is preinstalled on the seabed and then guides a connector hub on a riser into position. Again, this solution again does not allow for lateral or angular deviation.

[0012] Another challenge is how to connect a supported element or structure to the pile after the pile has been installed in the seabed. In this respect, a mechanical connector may be mounted to the top of a preinstalled pile but that additional subsea operation complicates installation. Alternatively, a mechanical connector may be integrated with a pile to be installed with the pile, but that makes the pile heavier, bulkier and more complex. For example, U.S. Pat. No. 9,140,384 describes attachment of a pipeline termination assembly to a subsea structure that has capture slots to receive a pivoting part of the termination assembly. As another example, U.S. Pat. No. 8,038,368 describes adding a structure to a pile foundation to receive trolleys and heads for initiation of pipeline laying. A lighter, simpler alternative to the top structure of the pile is required.

[0013] U.S. Pat. No. 7,794,177 discloses a stab and hinge-over pipeline end terminal assembly in which a socket receptacle on a pile receives an articulated pin or post of the pipeline end. That arrangement is simple and efficient in principle but is less effective in practice because it is challenging to insert the post into the receptacle on the pile. The top of the pile also requires reinforcement around the receptacle.

[0014] U.S. Pat. No. 8,425,154 shows a pile supporting a receptacle that includes a pair of vertical slots, into which pins of a frame that carries a new pipe section are lowered, to dock the frame with the receptacle. Once the pins are received in the slots, the initially vertical frame turns through a right angle to rest on a top surface of the pile. The new pipe section is then connected to an open end of an existing pipeline by a jumper. The precise positioning required to dock two pins simultaneously with respective slots is challenging in a marine environment, as water movement induces corresponding movement of the suspended frame relative to the fixed receptacle.

[0015] Against this background, the invention provides a subsea foundation that comprises: an upright pile; a support table on top of the pile; and a capture rod fixed to the pile and extending above a level of the support table. When the pile is installed at a seabed location, a lower end portion of the capture rod is embedded in the seabed.

[0016] The capture rod is preferably inclined away from a central longitudinal axis of the pile in an upward direction. Consequently, a lower end of the capture rod may be fixed to a side wall of the pile whereas the capture rod may be spaced from the pile at the level of the support table.

[0017] A strut cantilevered from the side wall of the pile may extend to the capture rod at a level above the lower end of the capture rod. Nevertheless, a free end portion of the capture rod suitably extends above the level of the support table.

[0018] The support table preferably widens in a direction away from the capture rod. More generally, the support table may be offset relative to the pile in a direction opposed to the capture rod. For example, a portion of the support table overhanging the pile may face away from the capture rod. In particular, a minor portion of the support table overhanging the pile may face toward the capture rod and a major portion of the support table overhanging the pile may face away from the capture rod. At least one stopper formation may project upwardly from an upper side of

the support table, the or each stopper formation preferably being offset toward the capture rod.

[0019] The inventive concept extends to a foundation system comprising the foundation of the invention and a connecting structure that is arranged to be seated on the support table of the foundation.

[0020] A catch structure joined to the connecting structure may be cooperable with the capture rod of the foundation. In particular, the catch structure may be pivotable with respect to the capture rod, for example about a substantially horizontal axis. It is also possible for the catch structure to be pivotable around the capture rod about a vertical axis, and for the catch structure to be slidable along the capture rod. Optionally, the catch structure is also pivotable relative to the connecting structure, for example about a horizontal axis

[0021] To allow for heave of an installation vessel, the catch structure preferably defines an elongate slot that is configured to accommodate the capture rod. For example, the slot may be defined by a loop formation of the catch structure. The slot and the capture rod may both lie in a common plane.

[0022] A levelling interface may be provided on an underside of the connecting structure to be sandwiched between the connecting structure and the support table.

[0023] The connecting structure may be joined to a line, that line being a pipeline, a mooring line or a riser, or a link to a pipeline, a mooring line or a riser. For example, the connecting structure may be a termination structure joined or linked to a pipeline, such as a PLET joined to an end of the pipeline. In other examples, the connecting structure may be an anchor structure that is joined or linked to a mooring line or a support structure that is joined or linked to a riser.

[0024] The inventive concept also embraces a corresponding method of connecting a line to a subsea foundation. The method comprises: suspending a connecting structure from the line; capturing the connecting structure on a capture rod that is fixed to a pile of the foundation and that extends above a level of a support table on top of the pile; and pivoting the connecting structure about the capture rod before landing the connecting structure on the support table.

[0025] The connecting structure may be pivoted about the capture rod around horizontal and/or vertical axes before being landed on the support table. The connecting structure may also be slid down the capture rod before being landed on the support table. Meanwhile, the connecting structure may be suspended between the capture rod and the line under tension in the line. It is also possible to pivot the connecting structure about the capture rod around a vertical axis after the connecting structure has been landed on the support table. Nevertheless, the connecting structure may be locked relative to the support table.

[0026] Conveniently, the capture rod may be engaged with a downwardly-projecting catch structure of the connecting structure, for example in a downwardly-elongate slot of the catch structure. The catch structure may be pivoted relative to the connecting structure, for example about a horizontal axis, before the connecting structure is landed on the support table.

[0027] After landing the connecting structure on the support table, movement of the connecting structure toward the capture rod may be restrained by engagement with at least one aforementioned stopper formation. Conversely, movement of the connecting structure away from the capture rod may be restrained by continued engagement with the capture rod after landing the connecting structure on the support table.

[0028] Embodiments of the invention provide an initiation and anchoring system that comprises an oblong and optionally articulated eye-link piece attached to or integrated into a termination structure such as a PLET (pipeline end termination). The eye link is hooked up for installation to an initiation/anchoring suction pile via a tilted pipe or rod that is fixed to the pile.

[0029] The foundation design of the invention provides excellent vertical tolerances while satisfying a requirement for the termination structure to be fully anchored with movement restricted along X, Y and Z axes. A stopper prevents movement of the termination structure in one direction, namely aft, while the oblong-eye link prevents movement of the termination structure in the

opposite direction, namely forward. Traditional PLET anchoring concepts employing a mooring chain, connected to a typical pile, cannot enable such extensive restriction of movement of the termination structure.

[0030] The invention enables quicker installation than traditional initiation methods with minimal ROV operations. There is no need to handle chains and no need for pre-initiation works such as pre-abandonment of chains.

[0031] The foundation design of the invention also provides a robust solution against degradation of the seabed soil. In particular, stability of the foundation is not significantly affected by soil erosion, for example due to long-term exposure to currents at the seabed. A simplified termination structure design is possible with no need for mudmats. The termination structure is instead stabilised on a support table on top of the pile.

[0032] The invention has good tolerance to variations in the pipeline heading relative to the as-installed heading of a pile. In particular, by virtue of the eye-link, the termination structure is free to rotate around the tilted rod during initiation. This eases initiation in a congested area of the seabed, particularly when compared to traditional anchoring/initiation systems. A multi-angle locking system may be engaged to lock the termination structure to the foundation after installation has been completed.

[0033] In comparison with the prior art, the invention provides an optimised seating area for landing and stabilisation of a termination structure. As there is a minimal area for a support table on top of a pile, the termination structure hinges over aft of the pile to optimise the landing area.

[0034] The invention simplifies the design of a pile. For example, the top deck of a prior art suction pile like that of U.S. Pat. No. 7,794,177 has to be reinforced to accommodate a stab-pin receptacle, but such reinforcement is unnecessary in a pile of the invention. The invention also allows the size of the pile to be reduced because anchoring forces are diverted toward the centre of gravity of the pile via the tilted rod.

[0035] The invention also simplifies installation procedures. A horizontal approach naturally engages the elongate eye-link with the tilted rod, with good tolerance to heave motion. This is in contrast to prior art arrangements like that of U.S. Pat. No. 7,794,177, which requires great precision to insert a stab-pin into a receptacle of a suction pile.

[0036] The overall sequence of a pipelay campaign employing the invention may be exemplified as follows. Firstly, a suction pile carrying a support table is fully installed within a designed heading tolerance. Post-installation, after a suitable interval, the pile is surveyed, for example using an ROV. In particular, heading and levelling data is collected and post-processed. Next, optionally, a levelling table is fabricated in accordance with the survey results and is then assembled with a PLET to complete fabrication of the PLET.

[0037] The offshore pipelay campaign can now begin. No pre-installation works are required. An ROV need only perform a general video inspection.

[0038] The PLET is deployed and lowered in a free-hanging configuration, with horizontal approach and hook-up of the eye-link to the tilted rod. Vertical tolerance allows for hook-up even with vertical excursion due to heave motion of the installation vessel.

[0039] After pivoting about a horizontal axis at the interface between the eye-link and the rod, the PLET is laid down on top of the support table within a designed heading tolerance. Pipelay is initiated and continues to stabilise initial layback of the pipeline before engaging a multiangle locking system to lock the PLET to the support table. A final inspection of the PLET and anchoring system is then performed.

[0040] Embodiments of the invention provide a foundation that comprises: a pile; a cantilever lateral extension to the pile; and a tilted connecting rod extending between a buried point of the pile and a distance above the mudline, via a point of the cantilever extension.

[0041] Embodiments of the invention also provide a foundation assembly comprising such a foundation and a connecting structure comprising a slot that is able to catch the connecting rod and

to slide along the connecting rod. The connecting structure may then be docked on and secured to the pile.

[0042] Embodiments of the invention also implement a method for connecting a line—such as a mooring line, a pipeline initiation line, a pipeline, or a link connected to a pipeline—to such a foundation. The method comprises: catching an upper tip of the connecting rod by a connecting structure that comprises an interface for the line; sliding the connecting structure down along the rod; and connecting the connecting structure to the top of the pile. The line may be connected to the connecting structure before or after the steps above.

[0043] The connecting structure may, for example, be connected to a pipeline initiation line, a mooring line or a pipeline touch-down point. The connecting structure may be a PLET.

[0044] The cantilever extension may be a rigid structure. The connecting structure may comprise a solid or rigid frame or a flexible lasso-like loop.

[0045] In summary, a subsea foundation of the invention comprises an upright pile, a support table on top of the pile, and a capture rod fixed to the pile and extending above a level of the support table. A lower end of the capture rod may be fixed to a side wall of the pile and the capture rod may be inclined away from a central longitudinal axis of the pile in an upward direction.

[0046] A line such as a pipeline or a mooring line is then connected to the subsea foundation by suspending a connecting structure from the line, capturing the connecting structure on the capture rod and pivoting the connecting structure about the capture rod before landing the connecting structure on the support table. The connecting structure may be pivoted about the capture rod around horizontal and/or vertical axes and may be slid down the capture rod before landing on the support table.

Description

[0047] In order that the invention may be more readily understood, reference will now be made, by way of example, to the accompanying drawings in which:

[0048] FIG. 1 is a schematic side view of a pile of the invention installed in the seabed;

[0049] FIG. 2 corresponds to FIG. 1 but shows a connecting structure of the invention in the form of a PLET, attached to the pile.

[0050] FIG. 3 is a top plan view of a pile of the invention with a PLET attached to it;

[0051] FIG. 4 is a perspective view of the pile and the PLET shown in FIG. 3;

[0052] FIGS. 5a and 5b are a sequence of end views of the PLET of FIG. 4 being lowered toward engagement with an upwardly-protruding capture rod of the pile;

[0053] FIGS. 6a to 6c are a corresponding sequence of side views of the PLET being lowered toward and into engagement with the capture rod of the pile;

[0054] FIGS. 7a to 7d are a sequence of side views of the PLET pivoting around and sliding down the capture rod into engagement with the top of the pile;

[0055] FIG. 8 is a schematic side view showing piles of the invention in use when mooring an FPSO;

[0056] FIG. 9 is a schematic side view showing a pile of the invention in use when controlling the touch-down point of a riser; and

[0057] FIGS. 10a to 10c are a sequence of schematic side views showing a pile of the invention in use when initiating a pipelaying operation.

[0058] Referring firstly to FIGS. 1 to 4 of the drawings, a subsea foundation 10 of the invention comprises an upright cylindrical pile 12, in this example a tubular suction pile, shown embedded in the seabed 14 following installation. In FIGS. 1 and 2, a rearward or aft direction is to the left and a forward direction is to the right, coinciding with the direction in which a pipeline or other elongate element anchored by the foundation 10 is to be laid.

[0059] The pile **12** is surmounted by a support table **16** that defines a support surface lying in a substantially horizontal plane. The support table **16** is asymmetric about a central longitudinal axis **18** of the pile **12**. Specifically, the support table **16** is offset forwardly such that a major overhanging portion **20** of the support table **16** extends forward from the pile **12**. Conversely, no more than a minor overhanging portion **22** of the support table **16** extends aft from the pile **12**.

[0060] At least one stopper formation **24** protrudes upwardly from the support table **16**. The or each stopper formation **24** is offset rearwardly toward an aft end of the support table **16**.

[0061] A connecting or capture rod **26** of the foundation **10** protrudes upwardly from the seabed **14** like a post, at a location rearward of an aft end of the support table **16**. The rod **26** joins the pile **12** below the level of the seabed **14**. In this example, the rod **26** is inclined relative to the vertical, extending upwardly and leaning rearwardly from an aft side of the pile **12**. Thus, the rod **26** is inclined away from the central longitudinal axis **18** moving upwardly from its junction with the pile **12**.

[0062] FIG. **4** shows that the rod **26** may be supported by a cantilevered strut **28** extending rearwardly from the aft side of the pile **12**, above the junction between the rod **26** and the pile **12**. The strut **28** is shown in FIG. **4** just beneath the seabed **14** but it could instead lie at or above the level of the seabed **14**. The rod **26** extends above the strut **28** to a free end at the upper and rearward extremity of the rod **26**.

[0063] The plan view of FIG. **3** shows that the support table **16** splays or widens forwardly from the rod **26**. In this example, the support table **16** has a shape approximating to an isosceles trapezium whose non-parallel sides converge rearwardly toward the rod **26**.

[0064] FIGS. **2** to **4** show a connecting structure, exemplified here by a PLET **30** at an end of a rigid pipe **32**, now engaged with the preinstalled foundation **10**. The PLET **30** rests on the support table **16**, where it is restrained against rearward movement relative to the pile **12** by the or each stopper formation **24** protruding upwardly from the support table **16**. FIG. **3** shows two stopper formations **24** whereas FIG. **4** shows a variant with a single, central stopper formation **24**. The or each stopper formation **24** may be positioned on the support table **16** before or after the PLET **30** is landed on the support table **16**.

[0065] Optionally, as shown in FIG. **2**, the PLET **30** has a levelling table or interface **34** on its underside to be interposed or sandwiched between the PLET **30** and the support table **16**. The interface **34** compensates for any significant departure from the horizontal of the support surface of the support table **16**, if a post-installation survey of the foundation **10** reveals such an anomaly. The interface **34** may, for example, be fabricated in response to metrology data obtained after installation and settling of the pile **12**.

[0066] A catch structure **36** extends rearwardly from the PLET **30**. The catch structure **36** could be rigidly attached to, or integral with, the PLET **30** but in this example the catch structure **36** is hinged to the PLET **30** about a pivot axis **38** that lies parallel to the base of the PLET **30** and so is horizontal. As seen in FIGS. **3** and **4**, the catch structure **36** comprises an open frame or eye link that defines a loop **40** capable of capturing and accommodating the rod **26** in a slot **42**. In this example, the loop **40** and hence the slot **42** defined by the loop **40** are elongate in the rearward direction. When the rod **26** is at the aft end of the slot **42**, the PLET **30** is restrained against forward movement relative to the pile **12**.

[0067] As best appreciated in FIG. **3**, the rod **26** when engaged in the slot **42** of the loop **40** serves as a pivot defining a vertical axis about which the PLET **30** and the pipe **32** can be displaced angularly. In this respect, the PLET **30** and the pipe **32** are shown displaced by an angle θ with respect to a central longitudinal plane **44**. The forwardly-widening support table **16** supports the PLET **30** throughout its range of angular movement.

[0068] When seated on the support table **16**, the PLET **30** is restrained against rearward movement by the or each stopper formation **24**, against forward and lateral movement by engagement of the rod **26** with slot **42** and against upward movement by the inclination of the rod **26**, noting that any

upward movement would also require rearward movement that is blocked. Thus, it will be apparent that the foundation **10** of the invention provides effective location of the PLET **30** along X, Y and Z axes while allowing pivotal yaw movement of the PLET **30** about the vertical or Z axis to accommodate variations in the heading of the PLET **30** and the pipe **32**.

[0069] Turning next to FIGS. **5a** to **6c**, these drawings show the PLET **30** suspended from the pipe **32** as the PLET **30** and the pipe **32** hang vertically in the water column while approaching the foundation **10** pre-installed in the seabed **14**. The front view of FIG. **5a** corresponds to the side view of FIG. **6a**, whereas the front view of FIG. **5b** corresponds to the side views of FIGS. **6b** and **6c**. In each case, the loop **40** of the catch structure **36** hangs beneath the PLET **30**, parallel to and in alignment with the pipe **32**. The rod **26** leans rearwardly toward the PLET **30**. The PLET **30** is moved downwardly and forwardly to bring the loop **40** into engagement with the tip of the rod **26**. [0070] Initially, as shown in FIGS. **5a** and **6a**, the loop **40** of the catch structure **36** is shown at a level above the rod **26** of the foundation **10**. Next, the PLET **30** is lowered as shown in FIGS. **5b** and **6b** to bring the loop **40** to the level of the top or free end of the rod **26**. Then, the PLET **30** is advanced forwardly toward the rod **26**, parallel to the seabed **14**, to engage the loop **40** around the tip of the rod **26** as shown in FIG. **6c**.

[0071] Advantageously, the vertical elongation of the slot **42** defined by the hanging loop **40** of the catch structure **36** provides for heave excursion if the pipe **32** and hence the PLET **30** experience vertical movement due to heave motion of the vessel that supports them.

[0072] When the loop **40** of the catch structure **36** has been engaged with the tip of the rod **26**, FIGS. **7a** to **7d** show how the PLET **30** pivots about a horizontal axis until the PLET **30** lies horizontally on the support table **16**. The horizontal axis about which the PLET **30** pivots, which is defined between the rod **26** and the aft end of the loop **40**, translates forwardly down the rod **26** as the loop **40** slides down the rod **26**. During this motion, the PLET **30** is suspended between the rod **26** and the pipe **32** by tension applied through the pipe **32** as a vessel (not shown) supporting the pipe **32** continues to move forwardly relative to the foundation **10**.

[0073] FIG. **7a** shows the PLET **30** starting to tilt from the vertical as the loop **40** of the catch structure **36** starts to slide forwardly down the rod **26**. In this example, the catch structure **36** pivots about the optional pivot axis **38** toward the horizontal to limit pivotal movement of the PLET **30**, thereby controlling bending stress in the pipe **32**. In this respect, it will be noted from FIG. **7b** that as the operation continues, the catch structure **36** pivots further toward the horizontal without requiring corresponding pivoting of the PLET **30**. In this way, the catch structure **36** decouples angular movement of the PLET **30** from engagement of the catch structure **36** with the rod **26**.

[0074] Eventually, as shown in FIG. **7c**, the PLET **30** pivots relative to the catch structure **36** to a more horizontal orientation as the adjoining end of the pipe **32** begins to approach the horizontal. Thus, the pipe **32** and the PLET **30** come back toward alignment with the catch structure **36** as the loop **40** of the catch structure **36** continues to slide down the rod **26**.

[0075] Finally, as the adjoining end of the pipe **32** reaches a horizontal orientation, the PLET **30** also reaches a horizontal orientation as it lands on the support table **16** of the foundation **10** as shown in FIG. **7d**. On landing on the support table **16**, a rearward end of the PLET **30** engages behind the or each stopper formation **24** to restrain the PLET **30** against rearward movement. Conversely, continued engagement of the rod **26** with the loop **40** of the catch structure **36** restrains the PLET **30** against forward movement.

[0076] Moving on now to FIG. **8**, this shows that the invention can be used with lines other than pipelines and with terminations other than PLETs. In this example, foundations **10** of the invention serve as anchors for catenary mooring lines **46** that fix the position of an installation such as an FPSO **48** floating at the surface **50**. Consequently, each mooring line **46** terminates at its bottom end in an anchor structure **52** that is coupled to the foundation **10** in the same manner as the PLET **30** shown in the preceding drawings. For this purpose, each anchor structure **52** has a catch structure **36** that engages the rod **26** of the foundation **10**. The catch structure **36** may comprise a

loop **40** like that shown in the preceding drawings and may be fixed or pivotable relative to the anchor structure **52**.

[0077] In a foundation **10** of the invention, the rod **26** leans away from the direction of tension in a pipeline or other line that is anchored by the foundation **10**. Thus, as tensile forces in the mooring lines **46** opposed about the FPSO **48** act in opposing directions, the rods **26** of the respective foundations **10** face in opposite outward directions with respect to the FPSO **48**. The foundations **10** and the associated mooring lines **46** are mirrored about a vertical plane intersecting the FPSO **48**.

[0078] FIG. **9** shows a foundation **10** of the invention being used to control the touch-down point **54** of a subsea riser **56**. In this example, the riser **56** is of wave configuration, comprising a hogbend section supported by buoyancy modules **58**. Here, a support structure **60** is landed on the foundation **10** in the manner of the PLET **30** and the anchor structure **52** of the preceding embodiments. The support structure **60** therefore has a similar catch structure **36** that engages the rod **26** of the foundation **10**. The support structure **60** also has a rearward extension **62** that encircles and engages the riser **56** to prevent movement of the touch-down point **54** across the seabed **14** away from the foundation **10**.

[0079] Turning finally to FIGS. **10a** to **10c**, these drawings show a foundation **10** of the invention being used to facilitate initiation of a pipeline **64** or other elongate element to be laid on the seabed. The pipeline **64** has a termination head **66** at one end that is landed on the foundation **10** in the manner of the PLET **30**, the anchor structure **52** and the support structure **60** of the preceding embodiments. Again, therefore, the termination head **66** has a similar catch structure **36** that is arranged to engage the rod **26** of the foundation **10**.

[0080] In FIG. **10a**, the termination head **66** has just been landed on the foundation **10**. The pipeline **64** is shown hanging as a catenary, initially held clear of the seabed **14** under tension applied by an installation vessel at the surface (not shown). The termination head **66** fixed to the foundation **10** reacts against hold-back tension applied to the pipeline **64** by the installation vessel.

[0081] In FIG. **10b**, the installation vessel has begun to lay the pipeline **64** on the seabed **14**. The remainder of the pipeline **64** hangs from the installation vessel as a catenary, held clear of the seabed **14** by tension. The termination head **66** fixed to the foundation **10** continues to react against hold-back tension in the pipeline **64**, as friction between the laid portion of the pipeline **64** and the seabed **14** is not yet sufficient to do so. However, when a great enough length of the pipeline **64** has been laid on the seabed **14** to react against hold-back tension applied by the installation vessel, the remainder of the pipeline **64** can be laid normally as shown in FIG. **10c**.

[0082] The pipeline **64** need not follow a straight path across the seabed **14**: the path may be curved in plan view, for example to create expansion loops, to follow favourable contours of the seabed **14** or to avoid other structures on the seabed **14**. In that case, the termination head **66** may continue to pivot about the capture rod **26** and slide on the support table **16** of the foundation **10** during installation of the pipeline **64**. Eventually, the termination head **66** may be locked to the support table **16** on an appropriate heading or may be left free to pivot and slide relative to the support table **16** in response to movement of the pipeline **64** in service.

[0083] Many other variations are possible without departing from the inventive concept. For example, the pipeline **64** shown in FIGS. **10a** to **10c** could be connected to the termination head indirectly via initiation rigging. Also, the invention can be used with rigid or flexible pipelines or with other elongate line elements such as umbilicals, cables or pipes for service fluids.

Claims

1. A subsea foundation, comprising: an upright pile; a support table on top of the pile; and a capture rod fixed to the pile and extending above a level of the support table; wherein when the pile is installed at a seabed location, a lower end portion of the capture rod is embedded in the seabed.

2. The foundation of claim 1, wherein the capture rod is inclined away from a central longitudinal axis of the pile in an upward direction.
3. The foundation of claim 1, wherein a lower end of the capture rod is fixed to a side wall of the pile.
4. The foundation of claim 1, wherein a strut cantilevered from the side wall of the pile extends to the capture rod above the lower end of the capture rod.
5. The foundation of claim 1, wherein a free end portion of the capture rod extends above the level of the support table.
6. The foundation of claim 1, wherein the capture rod is spaced from the pile at the level of the support table.
7. The foundation of claim 1, wherein the support table widens in a direction away from the capture rod.
8. The foundation of claim 1, wherein the support table is offset relative to the pile in a direction opposed to the capture rod.
9. The foundation of claim 8, wherein a portion of the support table overhanging the pile faces away from the capture rod.
10. The foundation of claim 9, wherein a minor portion of the support table overhanging the pile faces toward the capture rod and a major portion of the support table overhanging the pile faces away from the capture rod.
11. The foundation of claim 1, further comprising at least one stopper formation that projects upwardly from an upper side of the support table.
12. A foundation system comprising the foundation of claim 1 and a connecting structure arranged to be seated on the support table of the foundation.
13. The system of claim 12, further comprising a catch structure that is joined to the connecting structure and is cooperable with the capture rod of the foundation.
14. The system of claim 13, wherein the catch structure is pivotable with respect to the capture rod.
15. The system of claim 14, wherein the catch structure is pivotable about a horizontal axis with respect to the capture rod.
16. The system of claim 14, wherein the catch structure is pivotable around the capture rod about a vertical axis.
17. The system of claim 13, wherein the catch structure is slidable along the capture rod.
18. The system of claim 13, wherein the catch structure defines an elongate slot that is configured to accommodate the capture rod.
19. The system of claim 18, wherein the slot and the capture rod lie in a common plane.
20. The system of claim 19, wherein the slot is defined by a loop formation of the catch structure.
21. The system of claim 13, wherein the catch structure is pivotable relative to the connecting structure.
22. The system of claim 21, wherein the catch structure is pivotable relative to the connecting structure about a horizontal axis.
23. The system of claim 12, further comprising a levelling interface on an underside of the connecting structure, positioned to be sandwiched between the connecting structure and the support table.
24. The system of claim 12, wherein the connecting structure is joined to a line being a pipeline, a mooring line, a riser or a link to a pipeline, a mooring line or a riser.
25. The system of claim 24, wherein the connecting structure is a termination structure joined or linked to a pipeline.
26. The system of claim 25, wherein the connecting structure is a PLET joined to an end of the pipeline.
27. The system of claim 24, wherein the connecting structure is an anchor structure joined or linked to a mooring line.

- 28.** The system of claim 24, wherein the connecting structure is a support structure joined or linked to a riser.
- 29.** The system of claim 12, wherein when landed on the support table, the connecting structure is restrained by a stopper formation and by engagement of the catch structure with the capture rod.
- 30.** A method of connecting a line to a subsea foundation, the method comprising: suspending a connecting structure from the line; capturing the connecting structure on a capture rod that is fixed to a pile of the foundation and that extends above a level of a support table on top of the pile; and pivoting the connecting structure about the capture rod before landing the connecting structure on the support table.
- 31.** The method of claim 30, comprising pivoting the connecting structure about the capture rod around horizontal and/or vertical axes before landing the connecting structure on the support table.
- 32.** The method of claim 30, comprising sliding the connecting structure down the capture rod before landing the connecting structure on the support table.
- 33.** The method of claim 30, comprising suspending the connecting structure between the capture rod and the line under tension in the line before landing the connecting structure on the support table.
- 34.** The method of claim 30, comprising pivoting the connecting structure about the capture rod around a vertical axis after landing the connecting structure on the support table.
- 35.** The method of claim 34, followed by locking the connecting structure relative to the support table.
- 36.** The method of claim 30, comprising engaging the capture rod with a downwardly-projecting catch structure of the connecting structure.
- 37.** The method of claim 36, comprising pivoting the catch structure relative to the connecting structure before landing the connecting structure on the support table.
- 38.** The method of claim 37, wherein the catch structure is pivoted relative to the connecting structure about a horizontal axis.
- 39.** The method of claim 36, comprising engaging the capture rod in a downwardly-elongate slot of the catch structure.
- 40.** The method of claim 30, comprising restraining movement of the connecting structure toward the capture rod by engagement with at least one stopper formation after landing the connecting structure on the support table.
- 41.** The method of claim 30, comprising restraining movement of the connecting structure away from the capture rod by continued engagement with the capture rod after landing the connecting structure on the support table.
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